

Architecture Selection – Centralized Audit Trail Compliance Engine

We chose **Microservices Architecture** for the Centralized Audit Trail Compliance Engine.

The first reason for choosing Microservices Architecture is that the system contains several clearly separated responsibilities, including audit-log ingestion, JSON parsing, PII masking, searching, export generation, and integrity verification. Implementing these responsibilities as independent services improves modularity and allows each service to evolve independently.

The second reason is scalability. The system is expected to support complex queries across approximately 10 million audit records with a response time of under one second. A Microservices Architecture allows performance-intensive services such as the Search & Query Service to be scaled independently without requiring the entire system to be scaled.

From a security perspective, separating sensitive operations into individual services reduces the exposure of critical functionality. For example, PII masking can be performed by a dedicated service before audit data is stored, while integrity verification can be isolated from other application functions. This helps reduce the impact of a compromise in one service.

From a performance perspective, services such as ingestion, searching, and export generation can run independently and be optimized according to their workload. Frequently used services can also be scaled horizontally to handle higher loads without unnecessarily increasing the resources allocated to the entire application.

Therefore, Microservices Architecture provides an appropriate balance of scalability, modularity, security isolation, and performance for the Centralized Audit Trail Compliance Engine.